Algorithm 4: Label-setting algorithm with UB pruning, scoring, and early stopping

In this initial version, we use the following scoring formula:
$Score(L) = |C| + (1 - \tau)UB$; where $\tau = \frac{t - \text{start\_time}}{24{:}00}$

Input:

- V. Set of nodes
- E_s. Set of scheduled edges (v, u, dep, arr)
- E_t. Set of transfer edges (v, w, Δ)
- country(v). Country associated with node v
- $R(v, b)$. Precomputed reachable country sets
- bucket(t). Time bucket mapping function
- record. Current record to beat (e.g. 13)

Output:

- P. Sequence of labels representing a route within 24 hours

Initialize Labels$(v) \leftarrow \emptyset$ for all $v \in V$
Initialize priority queue $Q \leftarrow \emptyset$ // ordered by decreasing $score, |C|, UB, -t$
Initialize best_label $\leftarrow$ null
// Initialize labels at departures
for all $v \in V$ **do**
  for all scheduled departure from v at time dep with arrival arr **do**
    $L \leftarrow (v, dep, \{\text{country}(v)\}, \text{pred} = \text{null}, \text{start\_time} = dep)$
    Labels$(v) \leftarrow$ Labels$(v) \cup \{L\}$
    compute score(L)
    $Q \leftarrow Q \cup \{L\}$
// Main loop
while $Q \neq \emptyset$ **do**
  extract $L = (v, t, C, \text{pred}, \text{start\_time})$ with maximum score from Q
  // Enforce 24-hour journey limit
  if $t - \text{start\_time} > 24:00$ **then**
    continue
  // Upper bound computation
  $b \leftarrow \text{bucket}(t)$
  $\text{UB} \leftarrow |R(v, b) \setminus C|$
  // Record-based pruning
  if $|C| + \text{UB} \leq \text{record}$ **then**
    continue
  // Update best label
  if $\text{best\_label} = \text{null} \lor |C| > |\text{best\_label}.C|$ **then**
    best_label $\leftarrow L$
    if $|C| > \text{record}$ **then**
      break // early stopping: record beaten
  // Extend via scheduled edges
  for all $(v, u, dep, arr) \in E_s$ **do**
    if $dep \geq t \land dep - \text{start\_time} \leq 24:00$ **then**
      $t_{\text{new}} \leftarrow arr$
      if $t_{\text{new}} - \text{start\_time} > 24:00$ **then**
        continue

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            C′ ← C ∪ {country(u)}
            L′ ← (u, t_new, C′, pred = L, start_time = start_time)

            if ∄L″ ∈ Labels(u) such that
                (L″.t ≤ t_new ∧ L″.C ⊇ C′ ∧ (L″.t < t_new ∨ L″.C ≠ C′)) then
                remove all L″ ∈ Labels(u) such that
                    (t_new ≤ L″.t ∧ C′ ⊇ L″.C ∧ (t_new < L″.t ∨ C′ ≠ L″.C))
                Labels(u) ← Labels(u) ∪ {L′}
                compute score(L′)
                Q ← Q ∪ {L′}
    // Extend via transfers
    for all (v, w, Δ) ∈ E_t do
        t_new ← t + Δ
        if t_new − start_time > 24 : 00 then
            continue
        L′ ← (w, t_new, C, pred = L, start_time = start_time)

        if ∄L″ ∈ Labels(w) such that
            (L″.t ≤ t_new ∧ L″.C ⊇ C ∧ (L″.t < t_new ∨ L″.C ≠ C)) then
            remove all L″ ∈ Labels(w) such that
                (t_new ≤ L″.t ∧ C ⊇ L″.C ∧ (t_new < L″.t ∨ C ≠ L″.C))
            Labels(w) ← Labels(w) ∪ {L′}
            compute score(L′)
            Q ← Q ∪ {L′}
// Reconstruct path
P ← empty sequence
L ← best_label
while L ≠ null do
    prepend L to P
    L ← L.pred
return P
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